Subject Name: **Introduction to Algorithms (E1BCAT2)**

Model Question Paper 1

Section A: 2 Marks Questions

1. Define the term "algorithm."
2. What is the top-down design approach?
3. Write an algorithm to exchange the values of two variables.
4. What is program verification?
5. State the significance of analyzing an algorithm's efficiency.
6. Write the formula for the sine function computation.
7. What is a prime number? Give an example.
8. Define a binary search.
9. What is array partitioning?
10. Differentiate between linear search and binary search.

Section B: 5 Marks Questions

1. Explain the problem-solving aspects of computer programming.
2. Write an algorithm to reverse the digits of an integer.
3. Describe the process of finding the square root of a number using an iterative method.
4. Explain the method to generate the Fibonacci sequence with an example.
5. Write an algorithm to compute the greatest common divisor (GCD) of two integers.
6. Discuss the importance of removing duplicates from an ordered array and provide an algorithm for the same.
7. Compare selection sorting and insertion sorting methods.

Section C: 10 Marks Questions

1. Explain in detail the steps for top-down design and implementation of algorithms with an example.
2. Write an algorithm to compute the factorial of a number using recursion. Analyze its time complexity.
3. Describe the two-way merge process with an example. Write an algorithm for the same.
4. Write and explain an algorithm for binary search. Discuss its efficiency in comparison to linear search.

Model Question Paper 2

Section A: 2 Marks Questions

1. What is base conversion? Provide an example.
2. Define pseudo-random numbers.
3. Write an algorithm to find the smallest element in an array.
4. What is the importance of efficiency in algorithm design?
5. State the use of the "character to number conversion" algorithm.
6. What is the difference between array counting and array order reversal?
7. List the steps in a selection sort algorithm.
8. Define partitioning in the context of sorting.
9. What is meant by prime factorization?
10. Write the purpose of the two-way merge technique.

Section B: 5 Marks Questions

1. Write an algorithm to count the number of digits in an integer.
2. Explain the method to find the smallest divisor of an integer with an example.
3. Discuss the algorithm for generating prime numbers using the Sieve of Eratosthenes.
4. Write a program to remove duplicates from an ordered array. Explain the logic.
5. Compare sorting by selection and sorting by exchange.
6. Describe the process of finding the maximum number in a set of elements stored in an array.
7. Write an algorithm to sort an array using insertion sort and discuss its time complexity.

Section C: 10 Marks Questions

1. Explain with examples the various problem-solving aspects involved in designing algorithms.
2. Write and explain an algorithm to compute the Fibonacci number using recursion. Analyze its time and space complexity.
3. Explain the sorting by partitioning method with a detailed example.
4. Write and explain the steps for implementing a two-way merge algorithm. Discuss its advantages and disadvantages.

Model Question Paper 3

Section A: 2 Marks Questions

1. Define algorithm efficiency and state its types.
2. What is the purpose of array counting?
3. Write the steps to compute the sine of an angle.
4. What is character-to-number conversion? Provide an example.
5. Differentiate between insertion and exchange sorting techniques.
6. Define the smallest divisor algorithm.
7. What is the significance of merging in data handling?
8. Define the concept of array order reversal.
9. What is meant by raising a number to a large power? Provide an example.
10. Write the base condition for the Fibonacci number computation.

Section B: 5 Marks Questions

1. Write an algorithm to find the square root of a number using Newton-Raphson's method.
2. Explain the process of generating pseudo-random numbers with an example.
3. Write an algorithm to reverse the digits of an integer and explain it with an example.
4. Describe the binary search algorithm and compare it with linear search in terms of efficiency.
5. Write an algorithm to partition an array around a pivot element. Explain with an example.
6. Explain the importance of program verification with an example.
7. Describe the process of finding the prime factors of an integer using a step-by-step algorithm.

Section C: 10 Marks Questions

1. Explain the concept of top-down design in detail. Write a sample algorithm that follows this approach.
2. Write an algorithm to find the greatest common divisor (GCD) of two numbers using Euclid’s method. Analyze its efficiency.
3. Explain sorting by selection with a detailed step-by-step example. Write the algorithm and analyze its time complexity.
4. Write an algorithm for merging two sorted arrays and explain it in detail with an example. Discuss its application in real-world scenarios.

Government First Grade College, Honnavar
I Semester BCA Model Question Paper
Introduction to Algorithms

Time: 3 Hours **Max Marks: 80**

Instruction: Answer all the Sections.

SECTION – A

I. Answer any TEN questions. Each question carries TWO marks. (2X10=20)

1. Define Algorithm.
2. Define step – wise refinement.
3. What is program verification?
4. Write a simple strategy to find the smallest devisor of an integer.
5. What is the characteristic for prime factors of an integer?
6. How to choose parameter a, for generating pseudo random numbers?
7. State a problem for array counting method.
8. How to find the maximum number in a set?
9. Define searching & sorting.
10. What are the uses of sorting?

SECTION – B

II. Answer any SIX of the following. Each question carries FIVE marks. (5X6=30)

11. Explain sort by exchange with example.
12. Sort the following numbers using selection method. 20 10 30 15 5 25
13. Explain, how to reverse an array?
14. Explain the process of finding k^{th} smallest element from an unordered array.
15. Write an algorithm to find n^{th} Fibonacci number.
16. Write an algorithm to raising a number to a large power.
17. Explain the aspect of problem solving.
18. Write a note on order notation.

SECTION – C

III. Answer any THREE of the following. Each question carries TEN marks. (10X3=30)

19. Explain Top – down design process.
20. Explain the algorithm “Finding the square root of a number.”
21. How to remove a duplicate from an ordered array? Explain.
22. Explain quick sort with example.